Design of Hot and Ambient Air Mixing System and Air Supply System for Efficient Withering of Tea Leaves at St. Coomb Tea Factory, Tea Research Institute (TRI), Thalawakelle

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1. Project Overview

1.1 Introduction

Tea withering is a critical stage in tea manufacturing because it directly affects moisture reduction, leaf condition, aroma, colour, and final tea quality. At present, the withering process at St. Coomb Tea Factory is carried out using large open trays where green tea leaves are manually spread and turned by workers.

The existing system mainly depends on furnace hot air supplied through a conventional duct arrangement. However, the hot air is not properly mixed with ambient air before reaching the trays. This creates uneven air temperature and airflow distribution across the withering area. As a result, some tray sections receive excess heat while other sections receive insufficient airflow, causing uneven withering conditions.

The manual turning process also increases labour dependency, operator fatigue, and operational inconsistency. In addition, the absence of proper monitoring and control systems makes it difficult to maintain stable operating conditions throughout the process.

This project proposes an improved automated air mixing and air distribution system integrated with a mechanical leaf turning mechanism. The proposed system will use a controlled air mixing chamber, improved ducting arrangement, sensors, and basic instrumentation to provide uniform airflow and temperature distribution to all withering trays.

The project aims to improve tea quality, reduce withering time, minimize labour involvement, improve energy utilization, and modernize the existing withering process with minimum modifications to the factory layout.

1.2 Purpose and Need of the Project

The current tray withering process has several operational and quality-related limitations that reduce process efficiency and product consistency.

Major problems identified in the existing system are:

- Non-uniform mixing of hot air and ambient air.
- Uneven airflow distribution across trays.
- Temperature variations creating hot and cold regions.
- Inconsistent moisture removal from tea leaves.
- High labour requirement for manual leaf turning.
- Lack of real-time process monitoring and control.
- Increased withering duration and energy losses.

These issues directly affect tea leaf quality, process stability, labour cost, and factory productivity. Therefore, an improved engineered solution is necessary to achieve controlled and uniform withering conditions.

The proposed system will introduce better airflow management, automated leaf agitation, and basic instrumentation into the withering process. This will help improve operational efficiency while supporting modern tea manufacturing practices.

2. Analysis of Existing System

2.1 Existing System Description

The present withering system uses open tray withering beds supplied with furnace-generated hot air through a basic duct system. Tea leaves are manually turned every 1–2 hours to improve moisture removal.

The current system mainly depends on manual operation and operator experience. There is no dedicated air mixing chamber or automated airflow balancing arrangement.

Existing System Components

- Furnace hot air supply.
- Conventional air ducts.
- Open withering trays.
- Manual leaf turning process.
- Limited instrumentation.

2.2 Limitations of Existing System

Uneven Air Distribution

- The existing duct arrangement does not supply equal airflow to all trays, resulting in non-uniform withering.

Poor Temperature Uniformity

- Improper mixing between furnace hot air and ambient air creates temperature fluctuations within the withering area.

High Labour Dependency

- Frequent manual leaf turning increases labour cost and operator fatigue.

Inconsistent Tea Quality

- Uneven temperature and airflow conditions produce inconsistent moisture removal and leaf quality variations.

Lack of Process Control

- The current system does not include sensors or monitoring systems for temperature, humidity, or airflow.

Energy Inefficiency

- Uncontrolled air distribution causes unnecessary heat losses and inefficient energy utilization.

3. Project Objectives & Methodology

3.1 Project Objectives

The main objective of this project is to design an efficient hot and ambient air mixing and air supply system for improving tea leaf withering performance at St. Coomb Tea Factory.

Specific Objectives

1. To evaluate the existing tray withering system and identify problems related to airflow, temperature distribution, and moisture removal.
2. To design a controlled air mixing chamber for proper blending of furnace hot air and ambient air.
3. To develop an improved ducting system capable of providing uniform airflow and temperature distribution to all withering trays.
4. To design a mechanical leaf turning mechanism to automate tea leaf agitation.
5. To integrate temperature, humidity, and airflow sensors with a basic monitoring and control system.
6. To reduce labour dependency, improve process efficiency, and enhance tea quality consistency.

3.2. Proposed Methodology

The project will be completed through the following stages.

Stage 1 – Literature Review and Data Collection

A detailed study will be carried out on tea withering systems, industrial air mixing methods, duct design techniques, airflow control systems, and mechanical agitation mechanisms.

Operational data from the factory will be collected, including:

- Air temperature.
- Airflow conditions.
- Existing duct dimensions.
- Tray layout.
- Furnace operating conditions.

Stage 2 – Existing System Evaluation

The current withering arrangement will be analyzed to identify:

- Airflow imbalance.
- Temperature variation.
- Heat losses.
- Operational limitations.

Measurements and observations will be used to determine improvement requirements.

Stage 3 – Design of Air Mixing and Distribution System

An air mixing chamber will be designed using dampers, fans, and airflow balancing arrangements to achieve controlled mixing of hot air and ambient air.

The ducting system will be redesigned to:

- Deliver equal airflow to trays.
- Maintain uniform air velocity.
- Reduce pressure losses.
- Improve heat distribution.

Engineering calculations related to airflow rate, duct sizing, and pressure drop will be performed.

Stage 4 – Mechanical Leaf Turning Mechanism Design

A suitable mechanical mechanism will be designed to automate tea leaf turning.

The design process includes:

- Mechanism selection.
- Motor selection.
- Motion analysis.
- Structural design.
- Safety consideration.

The mechanism will be designed to minimize leaf damage while improving agitation efficiency.

Stage 5 – Instrumentation and Control Integration

The proposed system will include:

- Temperature sensors.
- Humidity sensors.
- Airflow sensors.
- Basic control panel.

Stage 6 – Simulation and Validation

Simulation and engineering analysis will be carried out to validate:

- Airflow distribution.
- Temperature uniformity.
- Mechanical performance.
- Overall system efficiency.

Stage 7 – Final Design and Documentation

Final CAD models, engineering drawings, calculations, and technical documentation will be prepared.

3.3 Proposed Software Tools

Software	Purpose
SolidWorks	3D modeling and mechanical design
AutoCAD	Technical drawings and layouts
ANSYS / CFD Software	Airflow and temperature analysis
MATLAB	Engineering calculations
Microsoft Excel	Data analysis and calculations

4. Project Planning and Timeline

Milestone 1 – Research and Existing System Study

Weeks 1–4

Activities:

- Literature review.
- Factory data collection.
- Existing system analysis.
- Problem identification.

Milestone 2 – System Design and Calculations

Weeks 5–7

Activities:

- Air mixing chamber design.
- Duct system design.
- Mechanical mechanism design.
- Engineering calculations.

Milestone 3 – Simulation and Development

Weeks 8–10

Activities:

- CFD analysis.
- Design validation.
- Instrumentation integration.
- Design improvements.

Milestone 4 – Finalization and Documentation

Weeks 10–14

Activities:

- Final design validation.
- Report preparation.
- Presentation preparation.
- Final documentation.

5. Impact Analysis

Economic Impact

The system can reduce labour cost, improve production efficiency, and minimize energy losses, leading to better operational performance.

Social Impact

Automation of leaf turning will reduce operator fatigue and improve workplace safety and ergonomics.

Environmental Impact

Efficient air mixing and controlled airflow distribution will reduce unnecessary energy consumption and improve sustainable manufacturing practices.

6. Expected Outcomes

The proposed project is expected to:

- Achieve uniform airflow and temperature distribution.
- Improve tea leaf withering consistency.
- Reduce withering time.
- Improve tea quality.
- Reduce labour requirement.
- Improve operator safety and ergonomics.
- Reduce energy losses.
- Modernize the current withering process.

7. Conclusion

This project focuses on improving the tea leaf withering process at St. Coomb Tea Factory through the design of an automated hot and ambient air mixing system integrated with an efficient air supply and mechanical leaf turning mechanism.

The proposed system addresses the major limitations of the existing process by improving airflow distribution, temperature uniformity, process monitoring, and operational efficiency. The integration of controlled air mixing, improved ducting, mechanical agitation, and instrumentation is expected to improve tea quality consistency while reducing labour dependency and energy losses.

The proposed design is technically feasible, economically beneficial, and suitable for practical implementation in the existing factory environment.